



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 28.07.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : DEZG612
Course Title : Materials Testing and Selection
Instructor-in-Charge : Srinivasan Periaswamy
Instructor(s) :
Tutorial/Practical Instructors:

- 1. Course Description:** Destructive testing: Tensile testing: Elastic modulus – Applications of elastic deformation - Case studies in modulus limited design; Strength and ductility – Case studies in yield limited design; Fracture – Case studies in fracture; Fatigue and Creep tests – Case studies in fatigue and creep failure. Corrosion tests – Types of corrosion – Case studies in wet corrosion. Non-destructive testing: Overview of the non-destructive testing; detection of surface flaws, detection of internal flaws, and Case studies. Materials selection in mechanical design - Materials properties charts – Basics of materials selection – Minimum mass design – Deflection limited design; Material selection for flywheels, springs, coupling, shaker tables and pressure vessels. Designing hybrid materials – Composites – Sandwich structures – Lattice structures. AI tools for materials selection. Lab: • Quasi-static and high-strain-rate testing of materials to determine their elastic and plastic properties. • Fatigue and Creep tests to determine the dynamic properties of materials • Experiments to determine the corrosion rate • non-destructive tests by DPT, MPI and Ultrasonic studies • Simulation studies on materials selection using ANSYS Granta Selector.

2. Scope and Objective of the Course:

3. Text Book(s)

T1	William D Calister Jr, "Material Science & Engineering, an Introduction", John Wiley & Sons, New York. 8th edition
T2	George E. Dieter, 'Mechanical Metallurgy', McGraw Hill, SI Metric Edition

Reference Book(s) & other resources

R1	William F. Smith, Javad Hashemi, 'Material Science & Engineering', In SI units Fourth edition, McGraw-Hill.
R2	Norman E. Dowling, Mechanical Behavior of Materials, Pearson Publications
R3	Marc Andre Meyers, Krishan Kumar Chawla, Mechanical Behavior of Engineering Materials Cambridge University Press (2009).
R4	Thomas H. Courtney, Mechanical Behavior of Materials, McGraw Hill Education
R5	Richard W. Hertzberg, Deformation and Fracture Mechanics of Engineering Materials, John Wiley & Sons
R6	Ravi Prakash, Non-Destructive Testing Techniques, New Age International Private Limited
R7	Ian Hutchings, Philip Shipway, Tribology: Friction and Wear of Engineering Materials, Butterworth-Heinemann publisher



R8	Joseph R. Davis, Corrosion: Understanding the Basics, ASM International
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5. Course Plan:

Module No.	Lecture Session	Reference	Learning outcomes
1-4	Structure property correlation of Engineering Materials	T1 (Ch1-4, Ch11-16), R1 (Ch1-3, 9-12), R2 (Ch3)T1	Understanding the relationship between the structure and properties of materials from atomic, crystal, micro and macro structure.
5-7	Structural characterization techniques	T1 (Ch3-4), R1 (Ch3-4), Lecture notes	Importance of materials characterization, Optical microscopy and Electron microscopy, grain size measurement, X-ray diffraction fundamental
8-11	Elastic-plastic behavior of materials - Review	T1 (Ch6), T2 (Ch2,3), R2 (Ch5-6), R3 (Ch2-3), R4,R5	Concept of stress and strain, Stress-strain relationship for elastic behavior, elastic and plastic behavior of Materials (metals, ceramics, polymers and composites)
11-14	Testing of materials- Tensile testing	T2 (Ch8), R2 (Ch4)	Tensile testing and test standards for metals, polymers and composites, Engineering stress strain curve, true stress strain curve, determination of mechanical properties from tensile test, Poisson's ratio, Instability in tension, Effect of temperature and strain rate.
15-20	Fracture Behavior of materials	T2 (Ch7, Ch11), R2 (Ch 7-8), R4 (Ch9-10)	Fundamentals of fracture, Types of fractures in metals, Theoretical cohesive



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			strength of metals, Theory of ductile and brittle fracture, Crack initiation and growth, stress concentration and fractures, Metallographic aspects of fracture, Fracture under combined stress, case studies. Impact testing and DBTT
21-25	Fatigue behavior of materials	T2(Ch12), R2 (Ch9-11, Ch14)	Fatigue of metals, Stress cycles, S-N curve, standards for fatigue test and determination of S-N curve, Theories of fatigue, Paris law, Factors affecting fatigue and mitigation methods, Origin of residual stress, Effect of residual stress, Residual stress measurement methods, Stress relief
26-30	Creep and Stress rupture	T2(Ch13), R2 (Ch 15), R3(Ch13)	High temperature behavior of materials, Creep curve, Creep and stress rupture test and test standards, Deformation at elevated temperature, Factors affecting creep behavior, Interpretation of creep data, Larson Miller Parameter
31-35	Non Destructive testing	R1(Ch7), R6	Overview of the Non-Destructive Testing, Detection of surface flaws, Detection of internal flaws, Visual inspection, Liquid Penetrant Testing, Magnetic Particle



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			Testing, Thermography, Eddy Current Testing, Radiography, Ultrasonic Testing, Acoustic emission
36-38	Introduction to wear & corrosion behavior of the materials	T1(Ch17), R1(Ch13), R3(Ch16), R7, R8	Basic understanding of Friction and wear, Laws of friction, Archard law's & equation, different type of wear, overview of Pin-on-disc wear test. Corrosion Electrochemistry, Nernst equation, forms of corrosion, Cathodic Protection
39-43	Selection of Materials and application	Ch.21 (T1)	Materials performance indicators. Selection of materials for flywheels, springs, coupling, shaker tables and pressure vessels

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 Min.	30%	<TEST_1>	Open book
Comprehensive Examination	3 h	35%	<TEST_C>	Closed Book
*Class participation and attendance	10 minutes	10%	Surprise quizzes (20 best out of 25)	Closed book
Lab experiments	2 hours/week	15%	Continuous	Open book
Project work	----	10%	Continuous	Open book

* This is applicable for students admitted after AY 2024-25.

7. Chamber Consultation Hour: Friday 5PM to 6PM (WILP Chamber in Library building)

8. Notices: All notices will be in the course page Nalanda

9. Make-up Policy: No make up for class quizzes

10. Attendance Policy: Each student must have at least 80% attendance

11. Minimum passing Marks criteria: At least 30 Marks/100 in total



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12. Note (if any): List of tentative experiments:

Sr No	Lab Details/ Project Details	Access
01	ImageJ open source software & UTM apk application developed by BITS-Pilani	Through online portal
02	Virtual Lab: ImagJ software: – 12 hours of practice & UTM application - 4 hours of practice – Graded for 20%	Through online portal
03	Specimen preparation and metallographic examinations to identify the specimens	Materials lab
04	Impact testing and DBTT	Materials Lab
05	Fatigue testing	Materials Lab
06	SEM analysis of Fracture surface	FSEM center
07	Tensile testing	Materials Lab
08	Corrosion Testing	Chemistry lab
09	Non Destructive testing	Central work shop

Prof.Srinivasan Periaswamy
Instructor-in-charge
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